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CONSTRAINTS ON THE TECTONIC EVOLUTION OF THE NAMAQUA PROVINCE III: PALAEOMAGNETIC AND $^{40}Ar/^{39}Ar$ RESULTS FROM THE GANNAKOURIEP DYKE SWARM

by

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ABSTRACT

Palaeomagnetic samples were collected from two dykes belonging to the Gannakouriep dyke swarm, one located in the centre of the Richtersveld Subprovince (8215), the other further south in the Steinkopf Domain (8212), where it has been clearly overprinted by a later metamorphic episode. Detailed thermal and A.F. demagnetization indicate that the primary remanence has been extensively overprinted in both dykes. The palaeomagnetic pole corresponding to the overprint directions are 80°N, 209°E ($n = 6$, $\kappa = 23$, $\alpha_{95} = 14$) for 8215 and 37°N, 353°E ($n = 6$, $\kappa = 72$, $\alpha_{95} = 8$) for 8212. Analyses of the thermal demagnetization data by using convergent remagnetization circles suggest a possible primary component with a corresponding palaeomagnetic pole 22°S, 67°E. $^{40}Ar/^{39}Ar$ step-heating analyses of a whole-rock sample of the 8212 yield a plateau spectrum with an integrated age of 542 ± 4 Ma; whereas, the $^{40}Ar/^{39}Ar$ spectrum for a whole-rock sample of 8215 yields a saddle-shaped spectrum with the minimum at ~400 Ma. The palaeomagnetic pole and the $^{40}Ar/^{39}Ar$ result for 8212 are both consistent with a metamorphic overprinting during the Pan-African Orogeny. The secondary remanence for 8215 appears to be a present-day field direction. The $^{40}Ar/^{39}Ar$ spectrum for 8215 may reflect partial ^{40}Ar loss during Karoo times, as suggested by $^{40}Ar/^{39}Ar$ results from the Vioolsdrif diorite. The palaeomagnetic pole for the suggested primary component does not coincide with previously published versions of the 500 to 900 Ma portion of the African apparent polar wander path, but does lie near the 700 Ma portions of proposed Gondwanaland APW paths.

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I. INTRODUCTION

McWilliams and Kröner (1981) showed that the 1,2 to 0,5 Ga old palaeomagnetic poles for the Congo and Kalahari shields (a total of 41 poles) could be incorporated into a single apparent polar wander (APW) path. From this observation, they derived an ensialic model for the origin of the Damara Orogen that did not require large-scale opening and closing of ocean basins, characteristic of a Wilson-cycle process. As they clearly indicated, however, most of the palaeomagnetic poles from the Kalahari Shield lie on the 1,2 to 1 Ga portion of their APW path; whereas, most of the palaeomagnetic poles from the Congo Shield lie on the 0,9 to 0,6 Ga portion. In fact, only a very few poles actually overlap from the two shields. Two of these poles were derived from a previously published palaeomagnetic study of the Gannakouriep dyke swarm (Piper, 1975a), which were referred to as the Klein Karas and the pre-Nama dykes. The ages of these poles were purportedly ~850 and ~650 Ma respectively, which coincide with the peak of the tectonic evolution of the Damara Orogen. The apparent agreement of these poles with poles from the Congo Shield is obviously of paramount importance to plate tectonic models for the origin of the Damara Orogen. For this reason, the authors undertook a detailed palaeomagnetic, petrographic, and $^{40}Ar/^{39}Ar$ study of two dykes belonging to the Gannakouriep dyke swarm with the goal of either verifying and improving upon the previously published results or refuting them.

II. GEOLOGICAL SETTING

The Gannakouriep dyke swarm, located in the south-western portion of the Kalahari Shield, was defined by Reid (1979) to include north to north-east-trending mafic dykes which predate the ~0,5 Ga Kuboos–Bremen granites, but which intrude the ~0,9 Ga Richtersveld Suite and the basal strata of the Gariep trough (Fig. 1).

The swarm is at least 300 km long and has a maximum width of ~150 km. Individual dykes typically exceed 5 m in width and some can be traced for distances of up to 100 km. The density of dyke intrusion increases from east to west and the trend progressively changes from north-east to north as the Gariep trough is approached (Fig. 1). The virtual absence of dykes in the Gariep trough to the west and in the basement rocks east of 18°E at 29°S suggests a genetic correlation between dyke emplacement and the early tectonic development of the Gariep trough. The swing in strike of the dyke swarm probably reflects tectonic re-orientation. North-trending dykes adjacent to the Gariep trough are deformed, faulted, and extensively recrystallized, whereas those further east and north-east are less deformed and relatively fresh.

The dykes grade from aphanitic chilled margins into coarse gabbroic interiors, which consist of subophitic intergrowth of plagioclase and clinopyroxene with patches of chlorite and minor carbonate enclosed in the subophitic network. Ilmenite and apatite are common accessory minerals. The dykes have a mild alkaline affinity

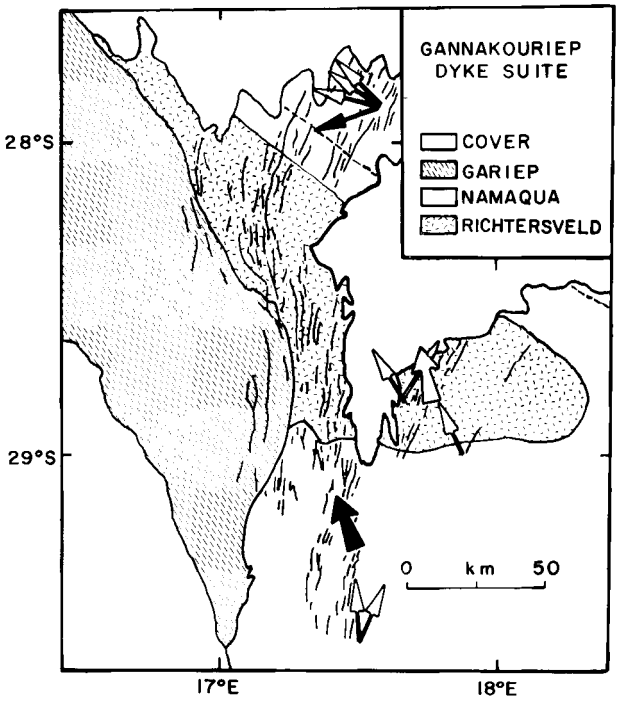


Figure 1
Geologic sketch map of the Gannakouriep dyke suite, adapted from Reid (1979), with the palaeomagnetic vectors plotted. The solid arrows represent positive inclinations, whereas the open arrows represent negative inclinations. The orientation of the arrow parallels the declination of the site average remanent magnetization. The small arrows are from Piper (1975a). The large arrows are this study.

compatible with a rifted continental environment (Hunter and Reid, 1985).

The Gannakouriep swarm is unconformably overlain by the sedimentary strata of the Nama Group (Fig. 1), which are older than 550 Ma (De Villiers and Burger, 1967) and younger than 720 Ma (Germs, 1972) or possibly 650 Ma (Köstlin, 1971).

Previous K-Ar and $^{40}Ar/^{39}Ar$ ages ranging from 1 200 to 200 Ma have been acknowledged (Reid, 1977), but not documented. Since some of the dolerite dykes, sheets, and sills in this area intrude Karoo strata, it was suggested by Reid (1977) that some of these ~200 Ma ages reported for the Gannakouriep, may actually represent Karoo dolerite. No final conclusions concerning the exact age of the Gannakouriep dyke swarm can be drawn from the radiometric age data presently available.

III. FIELD AND LABORATORY PROCEDURES

In 1982, with the assistance of the Precambrian Research Unit at the University of Cape Town, orientated samples were collected from two separate dykes from the Gannakouriep dyke swarm. Site 8212 was collected from a quarry on the Port Nolloth Road (Fig. 1), ~15 km west of Steinkopf. The strike of the dyke is 230°. Site 8215 was collected from a dyke striking 210°, located ~5 km south-east of Vioolsdrif. At this locality, a sample of the granite at the contact with the dyke was collected. Site 8215 is quite close to Piper's (1975a) sites 44 and 45, while site 8212 is close to Piper's (1975a) sites 40 and 41.

Specimens from all samples were subjected to detailed A.F. demagnetization (2,5, 5, 7,5, 10, 12,5, 15, 17,5, 20, 25, 30, 35, 40, 45, 50, 60, 70 mT), and detailed thermal demagnetization (200, 250, 300, 350, 400, 450, 500, 520, 540, 550, 560, 570, 580, 590, 600, 630, 660 °C). The data were treated with least squares techniques described by Onstott *et al.* (1986a). The average site statistics are reported in Table I, and the detailed demagnetization data in the Appendix. Polished thin sections from each site were examined under reflected and transmitted light and with magnetic colloid.

Two palaeomagnetic specimens which yielded the most stable palaeomagnetic results were selected for $^{40}Ar/^{39}Ar$ analyses. A 5 mm diameter by 5 mm long core was drilled from the centre of each specimen, wrapped in aluminium foil, and irradiated at the McMaster Nuclear Reactor Facility (Ontario, Canada). The samples were then stepwise degassed in 15 to 17 steps from 500 to 1 200 °C with a quartz furnace, and the argon analysed by a GD150 gas source mass spectrometer. Details of the technique are given in Onstott *et al.* (1986a). The analytical results are presented in Table II.

IV. PALAEOMAGNETIC AND PETROGRAPHIC RESULTS

The remanent intensity of samples from 8212 ranges from 0,199 to 0,021 Am^{-1} and the average median destructive field (MDF) is ~14 mT (Table I). Of the six samples, one behaves unstably with A.F. demagnetization, while the five remaining samples yield predominantly single component vectors, with a north-westerly, positive inclination direction (Fig. 2a). One sample (8212.04) reveals a hint that a higher coercivity component, with a south-easterly, positive inclination direction is present. This hint is reinforced by the thermal demagnetization data, which clearly indicate that the north-westerly, positive inclination component, stable to A.F. demagnetization, has a sharp unblocking temperature at 300 to 350 °C (Fig. 2b). Above this temperature range, the samples either behave unstably or yield a somewhat scattered south-easterly, positive inclination direction, with an unblocking temperature between 550 and 580 °C (Fig. 2b).

The remanent intensity of samples from 8215 is much weaker than 8212, ranging from 0,004 to 0,175 Am^{-1}, with an average MDF of 14 mT (Table I). Of the seven samples, six, including the granite contact rock, yield a northerly, negative inclination, single component direction, which is stable with A.F. demagnetization. One sample (8215.02) yields a stable, but anomalous westerly, horizontal direction (see Appendix). Only two samples (8215.04 and 8215.06) reveal any indication of a higher coercivity, second component (Fig. 2c). The northerly, negative inclination direction is unstable with thermal demagnetization (Fig. 2a), but has a distributed blocking temperature which makes isolation of the higher temperature component problematical. Above 540 °C, the remanence becomes unstable and drops sharply in intensity (Fig. 2d).

Analysis by means of converging remagnetization circles of the thermal demagnetization data for samples from both sites, using those samples which best exhibit two component behaviour (Fig. 3), tentatively suggests that the higher temperature component is easterly, with positive inclination. As seen in Fig. 3, the northerly, negative inclination direction from site 8215 is very close to the present-day field and axial dipole field directions.

Examination of polished thin sections of the two sites revealed the following:

1. dolerite from site 8215 has a very well-preserved primary ophitic texture, consisting of fresh pyroxene, altered, euhedral plagioclase, and altered opaque minerals. Some pyroxenes are rimmed by chlorite and actinolite. The opaque minerals are relic, high temperature exsolution intergrowths of ilmenite and titano-magnetite, except that the original titanomagnetite has been pervasively altered to cryptocrystalline sphene and extremely fine-grained (2 microns and less) hematite. The primary igneous ilmenite is still well preserved with minor inclusions and rims of hematite; and
2. metadolerite from site 8212 still maintains a relic

TABLE I
Palaeomagnetic Data from the Gannakouriep Dyke Swarm

Site No.	Site Lat.	Site Long.	N	n	Int.	MDF	Comp.	T	Dec.	Inc.	κ κ_1/κ_2	α_{95} α_1/α_2	Pole Long.	Pole Lat.
8212	−29,26	17,53	6	6	0,089	14	NW+	AF	338,9	35,7	72	7,9	352,5	36,7
				5			NW+	T	335,6	25,6	30	14,2	345,1	40,9
8215	−28,83	17,75	7	6	0,014	14	N−	AF	358,9	−56,4	23	14,1	208,8	80,9
				6			N−	T	6,9	−55,8	15	17,6	79,2	
			13	8			E+	T	93,1	63,9	−8/−1	7/11	66,8	−22,2

Notes:
N — number of samples collected.
n — number of samples included in site average statistics.
Int. — NRM intensity in Am^{-1}.
MDF — median destructive field in mT.
Comp. — components' classification.
T — treatment used in site average statistics, T is thermal and AF is A.F. demagnetization.
Dec., Inc. — vector average declination and inclination.
κ — Fisher's (1953) concentration parameter.
κ_1/κ_2 — Bingham's concentration parameter (Onstott, 1980).
α_{95} — Fisher's (1953) 95 % confidence angle.
α_1/α_2 — Bingham's 95 % confidence angles (Onstott, 1980).
Pole Long., Pole Lat. — VGP longitude and latitude.

TABLE II
$^{40}Ar/^{39}Ar$ Analytical Data for the Gannakouriep Dykes

T(°C)	f_1	f_2	Atmos. (%)	$^{40}Ar^*$ ($-10^{-6}cm^3$ STP)	Cum f(39)	$^{37}Ar_{Ca}/^{39}Ar_K$	Age(Ma)*
			Sample SAF82-12-6 (Whole Rock)				
525	−0,0040	0,0009	7,1	0,83	0,060	0,6	266 ± 1
550	−0,0153	0,0314	0,5	0,50	0,083	2,3	411 ± 1
600	−0,0036	0,0075	0,4	1,84	0,143	0,5	547 ± 2
650	−0,0030	0,0041	0,5	3,22	0,245	0,5	558 ± 2
700	−0,0033	0,0058	0,4	3,50	0,357	0,5	556 ± 1
750	−0,0091	0,0138	0,5	3,20	0,459	1,4	556 ± 1
800	−0,0191	0,0106	1,3	2,13	0,529	2,9	547 ± 1
830	−0,0388	0,0153	1,8	1,38	0,574	5,9	543 ± 1
860	−0,0427	0,0112	2,7	1,51	0,623	6,6	554 ± 1
890	−0,0363	0,0069	3,7	1,84	0,682	5,6	557 ± 1
920	−0,0280	0,0072	2,8	2,58	0,764	4,3	557 ± 1
950	−0,0219	0,0062	2,5	2,86	0,854	3,4	568 ± 2
1 000	−0,0143	0,0078	1,3	3,57	0,963	2,2	581 ± 1
1 050	−0,0612	0,0079	5,1	0,90	0,990	9,4	583 ± 3
1 100	−0,3729	0,0085	25,1	0,16	0,996	57,3	535 ± 8
1 150	−0,5602	0,0078	24,2	0,21	1,000	86,0	829 ± 12
Mass = 0,283 gm						Integrated Age =	541 ± 6
						Plateau Age (600–1 150 °C =	559 ± 6
			Sample SAF82-15-1 (Whole Rock)				
524	−0,1083	0,0002	52,4	0,36	0,008	16,6	1 138 ± 8
570	−0,0520	0,0004	49,6	0,18	0,018	7,9	513 ± 12
600	−0,0776	0,0011	36,6	0,21	0,031	11,9	477 ± 9
630	−0,0092	0,0002	35,8	0,24	0,049	1,4	422 ± 6
660	−0,0390	0,0009	29,9	0,35	0,076	6,0	399 ± 5
690	−0,0293	0,0010	23,3	0,33	0,100	4,5	406 ± 3
720	−0,0512	0,0032	13,9	0,39	0,128	7,9	427 ± 5
750	−0,0271	0,0059	4,5	0,38	0,156	4,2	419 ± 3
800	−0,0103	0,0019	5,3	0,76	0,212	1,6	415 ± 2
850	−0,057	0,0010	5,4	2,18	0,375	0,9	406 ± 1
900	−0,0070	0,0012	5,3	3,60	0,614	1,1	452 ± 1
950	−0,0090	0,0012	6,9	2,06	0,753	1,4	445 ± 1
975	−0,0144	0,0023	6,0	0,74	0,807	2,2	411 ± 2
1 000	−0,0162	0,0020	8,2	0,58	0,851	2,5	395 ± 3
1 025	−0,0172	0,0017	10,0	0,48	0,890	2,6	382 ± 3
1 050	−0,0232	0,0013	15,7	0,54	0,931	3,6	395 ± 2
1 075	−0,0597	0,0031	15,8	0,42	0,961	9,2	425 ± 4
1 100	−0,0432	0,0013	21,6	0,42	0,987	6,6	481 ± 4
1 150	−0,1882	0,0038	24,1	0,27	1,000	28,9	619 ± 8
Mass = 0,307 gm						Integrated Age =	437 ± 6
Standard J-value = 0,02318 ± 0,00032*							

*Weighted mean of four fusions of irradiation standard Mmhb-1, which has an age of 519,4 Ma (Alexander *et al.*, 1978).
Notes:
$f_1 = 1/[1-(36/39)_{Ca}/(37/39)_M)]$, and $f_2 = f_1[1-(36/39)_{Ca}/(36/39)_M]$, where $(\ )_{Ca}$ = isotope ratio of argon extracted from irradiated calcium salts and $(\ )_M$ = isotope ratio of argon extracted from irradiated unknown.
f(39) = cumulative fraction of ^{39}Ar released in each step. Age (Ma) — the date calculated using the following decay constants:
$\lambda\varepsilon = 0{,}581 \times 10^{-10}\ yr^{-1}$; $\lambda_\beta = 4{,}961 \times 10^{-10}\ yr^{-1}$; $^{40}K/K = 0{,}0116$ atom per cent (Steiger and Jäger, 1977). The quoted error is one standard deviation and not included the error in the J-value or the standard error.
Integrated Age — the age and error calculated from the sum total gas from all steps; the error includes the error in J-value.
Plateau Age — the age and error calculated from the sum total gas from those steps, the ages of which fall within 2 S.D. of each other; the error includes the error in J-value. Mass — milligrams.

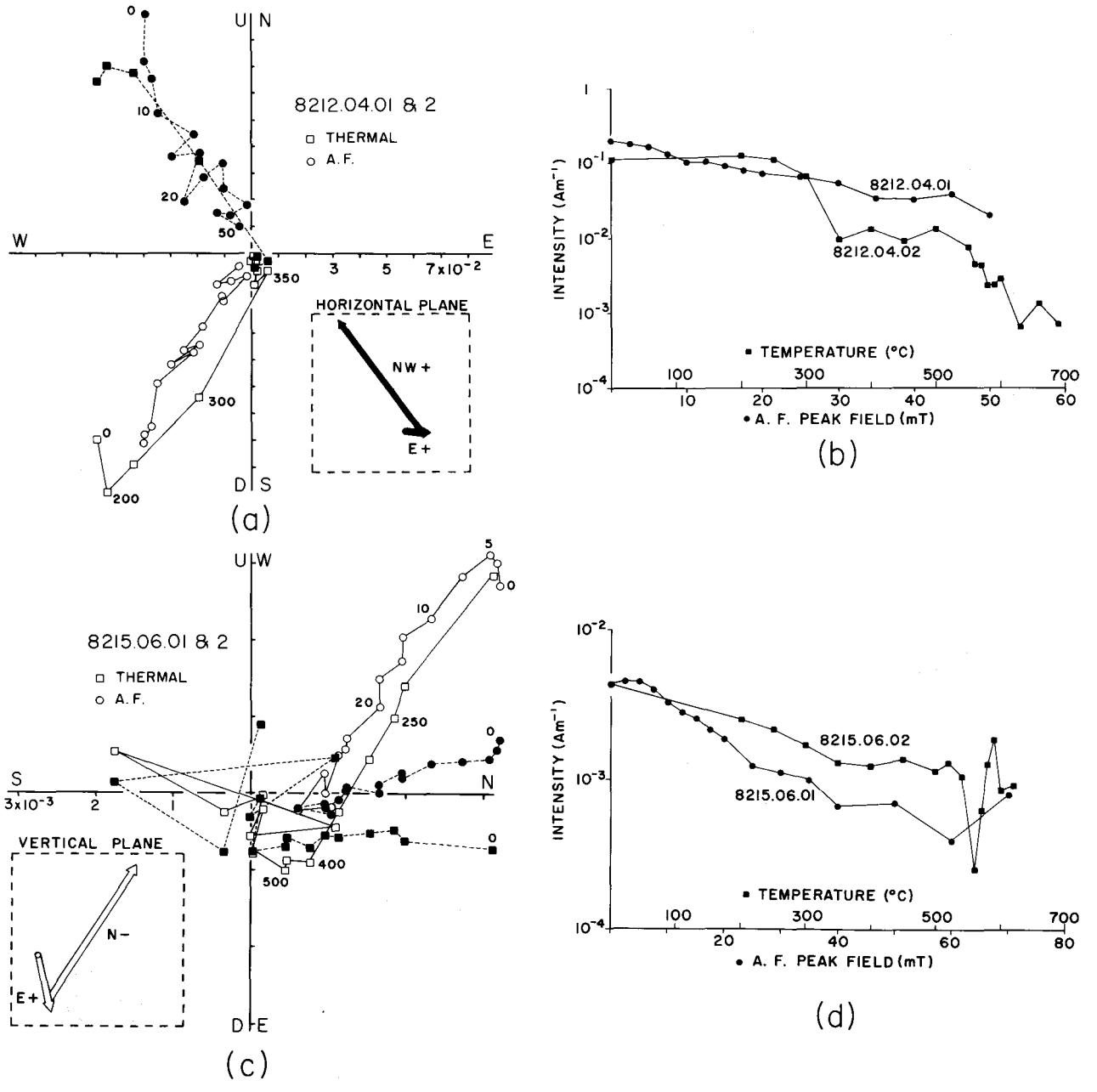


Figure 2

(a) Orthogonal plot of A.F. (circles) and thermal (squares) demagnetization of two specimens from sample 8212.04.01 and 02. The horizontal projection is plotted with solid symbols and dashed lines, whereas the vertical projection is plotted with open symbols with solid lines. (b) Orthogonal plot of A.F. and thermal demagnetization of sample 8215.06.01 and 02, with the horizontal projection plane rotated 90° CW for clarity. Symbols are the same as (a). Intensity is in S.I. units (Am^{-1}).

primary ophitic texture, but the pyroxenes are somewhat more altered, and sericitization of the plagioclase is intense. The relic, high temperature exsolution textures of the opaques are less well preserved, but, as in 8215, the original titanomagnetite has been completely replaced by cryptocrystalline sphene and the ilmenite is relatively unaltered. Unlike 8215, however, fine-grained pyrrhotite and pyrite occur with the cryptocrystalline sphene, replacing the titanomagnetite, and hematite is completely absent.

V. $^{40}Ar/^{39}Ar$ RESULTS

The $^{40}Ar/^{39}Ar$ step-heating analyses of the whole-rock sample from site 8212 (Table II) yield a good plateau spectrum (Fig. 4), with an integrated age of 541 ± 4 Ma, a plateau age (600–1 100 °C steps) of 542 ± 4 Ma. When plotted on a 40/36 *v.* 39/36 diagram (Fig. 5a), the data yield an errorchron age of 547 ± 226 Ma, and when plotted on a 36/40 *v.* 39/40 plot, the data yield an intercept age of 534 ± 5 Ma (Fig. 5b). This age is consistent with a 532 ± 10 Ma K-Ar age reported in Kröner (1974, C.J.D. Adams, pers. comm.) and a 543 ± 15 Ma K-Ar age reported in De Villiers (1969, by N.J. Snelling) for other Gannakouriep dykes.

The $^{40}Ar/^{39}Ar$ step-heating analyses of the whole-rock samples from site 8215 (Table II), however, yields a saddle-shaped spectrum (Fig. 4) characteristic of excess ^{40}Ar contamination (Lanphere and Dalrymple, 1976), with a meaningless integrated age of 428 ± 3 Ma. Although the minimum in the saddle is usually taken as the maximum estimate for the age of the dyke (Lanphere and Dalrymple, 1976), the ~400 Ma minimum given here is clearly too young for intrusion age, given the previously discussed geological constraints. When plotted on a 40/36 *v.* 39/36 diagram, the data yield an errorchron age of 393 ± 172 Ma (Fig. 6a). When plotted on a 36/40 *v.* 39/40 diagram (Fig. 6b), the excess ^{40}Ar

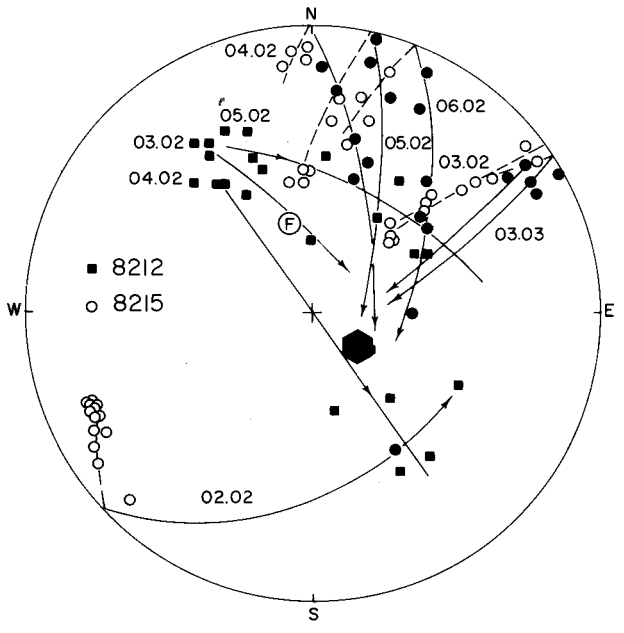


Figure 3
Equal-area plot of the thermal demagnetization data for samples from sites 8212 (squares) and 8215 (circles). Open symbols are upper hemisphere and closed symbols are lower hemisphere. Also plotted of the projections of the least-squares fitted remagnetization planes (solid lines are lower hemisphere, and dashed lines are upper hemisphere). "F" is the present-day field direction and the large solid hexagon marks the convergence direction of the remagnetization circles.

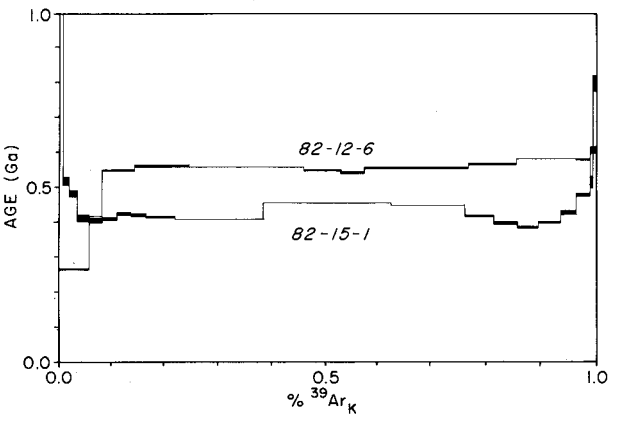


Figure 4
$^{40}Ar/^{39}Ar$ spectra for whole-rock samples of 8212.06.01 and 8215.01.01.

steps are clearly discernible and when excluded, the remaining data yield an intercept age of 411 ± 5 Ma. This age is identical within error to a 409 ± 16 Ma conventional K-Ar whole-rock age for site 42 of Piper's (1975a) pre-Nama dykes, quoted by Piper as a personal communication from D.C. Rex (1973).

In both samples, the majority of the argon probably resides in the altered feldspar. In order to estimate the retentivity of this whole rock for argon, the writers have applied the thermochronometric technique of Berger and York (1981), which determines the argon closure temperature from the $^{40}Ar/^{39}Ar$ step-heating data. A

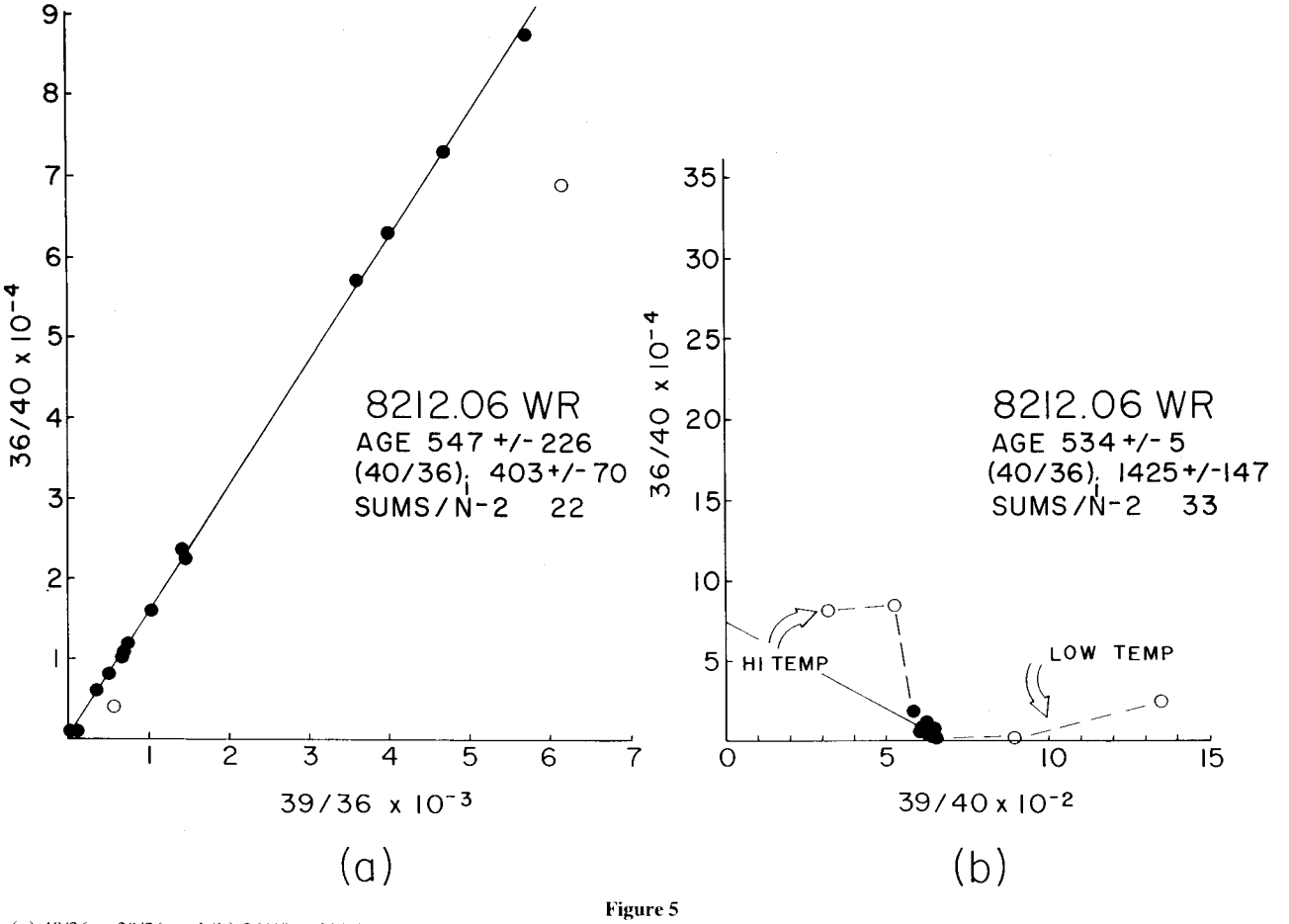


Figure 5
(a) 40/36 *v.* 39/36 and (b) 36/40 *v.* 39/40 plot for the whole-rock sample, 8212.06.01. Open arrows identify those temperature steps that have lost significant radiogenic argon and were not included in the least squares fit (open symbols are those data points excluded).

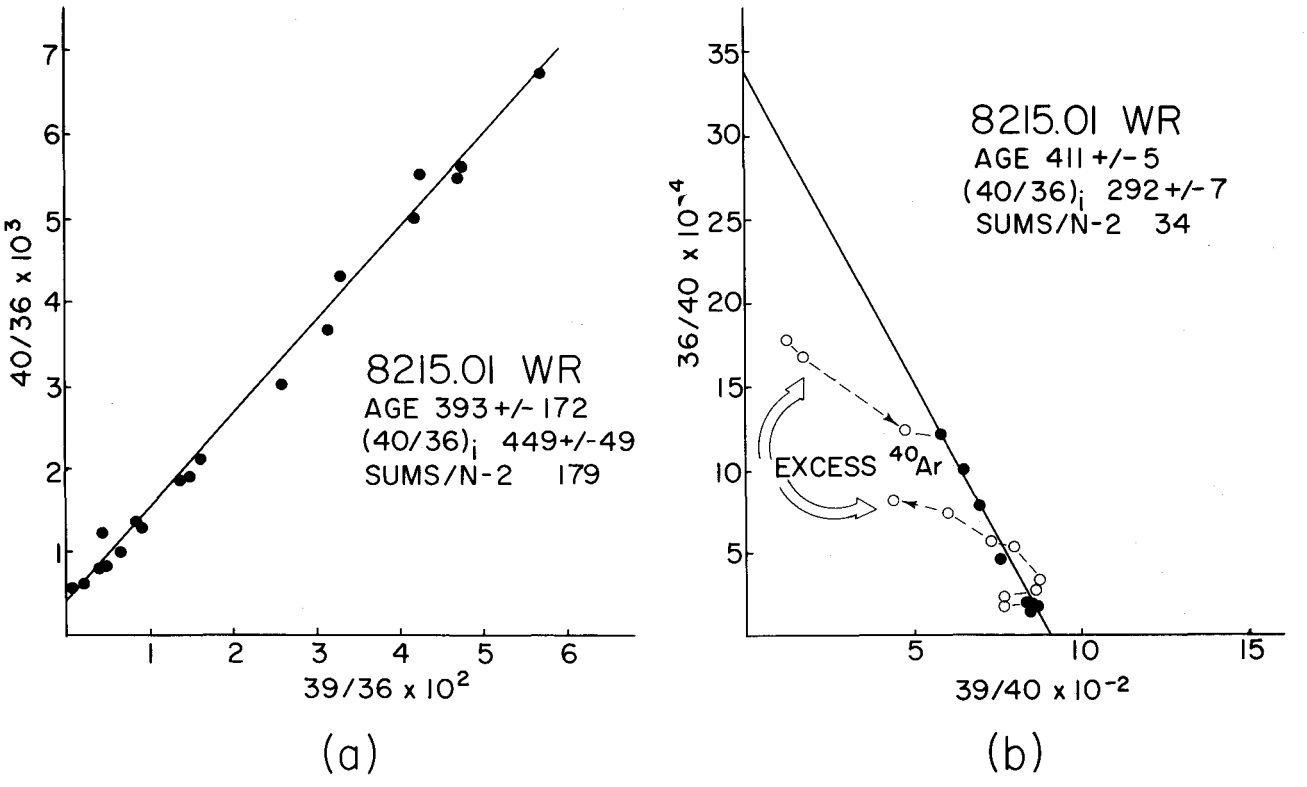


Figure 6
(a) 40/36 *v.* 39/36 and (b) 36/40 *v.* 39/40 plot for the whole-rock sample, 8215.01.01. Open arrows identify those temperature steps contaminated by excess argon and not included in the least squares fit (open symbols are those data points excluded).

closure temperature of 213° ± 71 °C was obtained for the sample from 8212 and a temperature of 233° ± 6 °C was obtained for the sample from 8215, both for a cooling rate of 5 °C/Ma. Since this approach works best for pure separates of anhydrous minerals (i.e. feldspar), these results should only be considered approximations.

VI. DISCUSSION

The presence of pyrrhotite in 8212 and hematite in 8215 and the complete absence of any observable magnetite in both dykes, probably accounts for the low intensities of their magnetizations, generally less than 0,1 Am^{-1}, the lesser intensity of 8215 compared to 8212, and the fairly high MDF values. The observed presence of pyrrhotite in 8212 is entirely consistent with the sharp unblocking temperature between 300 and 350 °C, as the Neel temperature for pyrrhotite is ~320 °C (Craig and Scott, 1974). The observed presence of hematite with a wide range of grain sizes is consistent with both the distributed blocking temperature and the present-day field direction overprint, as fine-grained hematite is notorious for carrying a viscous magnetization (Dunlop and Stirling, 1977). The preferential replacement of titanomagnetite by cryptocrystalline sphene has been described elsewhere (Haggerty, 1976), but its origin is debatable. It may represent a late-stage, post oxy-exsolution, metasomatic alteration or it may record a low-grade metamorphic episode. In either case, the difference between the observed assemblages in 8212 and 8215, can be explained by variation in the fS_2 (higher in 8212).

For the pre-Nama dykes, Piper (1975a) reported the presence of "pyrite and large, generally unoxidized, titanomagnetites intersertal with respect to the silicates" and titanomagnetites present as "fine acicular grains in the groundmass which appear to be largely unoxidised". These observations would seem to contradict those reported here, unless it is assumed that Piper (1975a) mistook the ilmenite, which is fresh and relatively unoxidised, for titanomagnetite. The ilmenite exsolution lamellae lying within the cryptocrystalline sphene alteration might have been mistaken for titanomagnetite lathes in a groundmass. For the Klein Karas dykes, Piper (1975a) reported the presence of titanomagnetite "subjected to extensive deuteric alteration with dense exsolution of ilmenite lamellae. The magnetite remains fresh, the ilmenite lamellae are variably altered to aggregates of rutile and haematite". These observations are consistent with those reported here if, again, Piper has mistaken ilmenite for titanomagnetite and the rutile hematite alteration for the cryptocrystalline sphene and hematite alteration. This would seem especially likely since the remanent NRM intensities reported by Piper (1975a) for both sets of dykes are very weak, on the order of 0,04 to 0,09 Am^{-1}, but consistent with the values reported here (Table I).

The 540 Ma $^{40}Ar/^{39}Ar$ date recovered from the whole-rock sample of 8212 clearly indicates that the dyke has been subjected to a low-grade thermal episode, probably associated with the Pan-African Orogeny. Consequently, the low-temperature remanence of 8212 could represent a thermochemical remagnetization at this time. Based upon closure temperature calculations presented in the previous section, the whole-rock argon isotopic system could have been reset by a thermal pulse with a minimum peak temperature between 200 and 250 °C. To thermo-viscously reset any prior magnetization carried by the pyrrhotite in 8212 would require peak temperatures of 290 to 320 °C. Since there are no isotopic constraints on the maximum temperature of the Pan-African orogenic effect, the secondary magnetization could have been acquired as a thermo-viscous overprint.

Regardless of the specific mode of origin, the palaeomagnetic pole corresponding to this low temperature magnetization falls off the coast of north-west Africa (Fig. 8) within a cluster of 500 to 600 Ma poles from the various cratons that make up Gondwanaland (listed in Table III). Included among these poles are results derived from late or

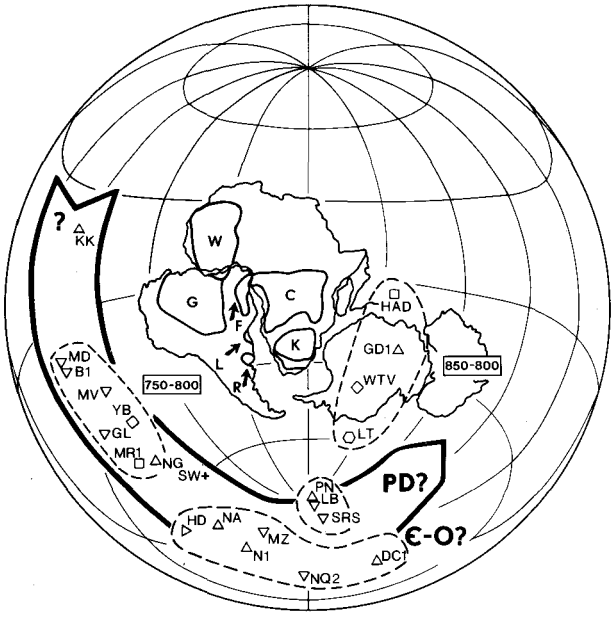


Figure 7

Equal area stereo projection showing the 1,3 to 0,9 Ga palaeomagnetic poles for the following Gondwanaland shields: South American (Sao Francisco) Shield (hexagon); Kaapvaal Craton and Namaqua Province (upward pointing triangles); Congo Shield (downward pointing triangles); Indian (Dharwar and Arravali) Shield (squares); Australian (Western, Northern, and Southeastern) Shield (diamonds). Abbreviations correspond to those given in Table III. The following individual cratons are delineated with thick lines; the Guiana Shield (G); the Sao Francisco Craton (F); the Luis Alvez Craton (L); the Rio de Plata Craton (R); the West African Shield (W); the Congo Shield (C); and the Kaapvaal Craton (K). The APW path for Gondwanaland for this time period (Klootwijk, 1980c) is portrayed with the thickest lines. Dashed lines have been used to delineate four clusters; and 750–800 Ma cluster, a 800–850 Ma cluster, a present-day overprint cluster (PD?), and a cluster which could be ~650 Ma or could represent Cambrian–Ordivician overprints (ε-O?). Continents and corresponding palaeomagnetic poles have been restored using a Smith and Hallam (1970) reconstruction, with Africa kept in its present-day position.



Figure 8

Equal area stereo projection showing 0,6 to 0,45 Ga palaeomagnetic poles for the following Gondwanaland shields: South American Shield (hexagons); the Kalahari Shield (upward pointing triangles); the Congo Shield (downward pointing triangles); the Arabian–Nubian Shield (rightward pointing triangles); the West African Shield (leftward pointing triangles); the Indian Shield (squares); the Australian Shield (diamonds); and the East Antarctica Shield (circles). Solid symbols are poles, whereas open symbols are antipoles. The abbreviations correspond to those given in Table III. Continents and corresponding palaeomagnetic poles have been restored using a Smith and Hallam (1970) reconstruction, with Africa kept in its present-day position. The following Precambrian cratons have been delineated with thick lines: the Dharwar Craton (D); the Arravali Craton (A); the Singhbhum Craton (S); the Western Australia Shield (Pilbara and Yilgarn cratons) (Y); the Northern Australian Shield (Kimberly, Mt Isa, Arunta, and Musgrave blocks and Amadeus Basin) (A); and the Southeastern Australian Shield (Gawler, Willyama, and Mt Painter blocks) (g).

post-tectonic igneous rocks from within Pan-African mobile belts, such as the Ntonya Complex (523–556 Ma, NR in Fig. 8) of the Moçambique Belt, the Damaran granites (450–550 Ma, DG in Fig. 8), the Adma diorite (~590 Ma, ADD in Fig. 8), and the Hook intrusives (519–472 Ma, HI in Fig. 8). All together, these data indicate that by 550 Ma or Early Cambrian, any large-scale (≥1 000 km) tectonic convergence or displacements in the Pan-African system had ceased.

The anomalously young, 411 Ma date for 8215 is understandable in light of the 342 Ma $^{40}Ar/^{39}Ar$ date for plagioclase from the Vioolsdrif diorite (Onstott *et al.*, 1985a). This result suggests that a young, low temperature event, probably Karoo in age, has caused partial loss of ^{40}Ar from the plagioclases in the Vioolsdrif area where 8215 was collected. This may also account for the ~200 Ma $^{40}Ar/^{39}Ar$ dates obtained for Gannakouriep dykes by S.R. Charlton (mentioned in Piper, 1975a as a personal communication). $^{40}Ar/^{39}Ar$ analyses of feldspars from the Namaqua Province (Onstott *et al.*, 1986b), on the other hand, reveal no indications of any thermal event younger than the Pan-African episode. This is consistent with the ~540 Ma date of 8212, since the majority of the argon probably resides in the altered plagioclase feldspar recrystallized during Pan African metamorphism.

The northerly, negative inclination component (Table I) present in 8215, which is stable to A.F. demagnetization, but unstable to thermal demagnetization, yields a direction and a palaeomagnetic pole not significantly different from the present-day geographic pole. Using the thermal demagnetization data from 8215.06.02, and assuming this secondary component is carried by hematite, then the calculated magnetic blocking temperature (York, 1978; Dodson and McClelland-Brown, 1980) for that component ranges between 50 and 85 °C. Such low temperatures are consistent with the hypothesis that the direction represents a viscous overprint and does not record a Precambrian field direction.

The palaeomagnetic pole derived from the secondary magnetizations of site 8215 coincides with the pre-Nama dyke pole reported by Piper (1975a) (PN in Fig. 7). The age quoted by Piper for his pre-Nama dyke pole is based upon "The pre-Nama dykes west of Vioolsdrif that were dated by Köstlin (1971) at 653 ± 70 My (Piper, 1975a). The "pre-Nama dykes" referred to by Piper in this case are actually *felsic* dykes that also underlie the Nama Group, but are not necessarily related to the *mafic* Gannakouriep dyke swarm sampled by Piper. Furthermore, the Rb-Sr data reported by Köstlin (1971) and more recently by Allsopp *et al.* (1979) for these felsic dykes define a 490 ± 30 Ma average date of isochrons which is similar to the $^{40}Ar/^{39}Ar$ results reported in this study and probably represents the time of resetting. Assuming an initial ratio of 0,705 and a closed system evolution and using mean Rb/Sr ratios, Allsopp *et al.* (1979) inferred a 676 ± 124 Ma age for intrusion of these *felsic* dykes. Nevertheless, this age was taken for the age of the pre-Nama dyke pole (Piper, 1975a), which was then utilized

TABLE III
0,5 to 0,85 Ga Palaeomagnetic Data for the Gondwanaland Continents

Shield Formation	Abbv.[a]	No. of Sites (samples)	Lat. (°N)	Long. (°E)	K (κ)	A_{95} (α_{95})	Possible Magnetization Age	Refs.
East Antarctica Shield								
Mirnyy Station Charnockite	MSC	5	2	208	(25)	(16)	520 ± 24 R	1
Sr. Rondane Intrusives	SRI	3	28	190	—	(5)	489–503 R bio, 528 R mu 510–540 U	2
Taylor Valley Lamproite	TVL	4	9	207	—	(11)	~470	3
Lutzow-Holm Bay Gneiss	LHG	5	19	203	—	—	Cm.-O S	4
Aravalli Craton, India								
Upper Bhander Sandstone	BS1	6	48,5	213,5	(137)	(6)	Up.Pc–Lw.Cm. S	5
Upper Bhander Sandstone	BS2	3	31	199	(200)	(6)	Up.Pc–Lw.Cm. S	6
Bhander Sandstone	BS3		51,5	222,5			Up.Pc–Lw.Cm. S	7
Upper Rewa Sandstone	RS1	2	35	222	(15)	(14)	Up.Pc S	6
Upper Rewa Sandstone	RS2		45	191,5			Up.Pc S	7
Purple Sandstone Salt Range.	PSR	(10)	28	212	(15)	(12)	Lw.Cm. S	8
Salt Pseudomorph Beds	SPB	6	27	213,5	(129)	(6)	M.Cm. S	9
Malani Rhyolite	MR1	8	80,5	43,5	(35)	(8)	729 ± 15 R	10, 12
Malani Rhyolite	MR2		80	45			729 ± 15 R	12, 11
Kaimur Sandstone	KS	20	82	286	—	(6)	>1 116 ± 12 R ph	13, 11
Dharwar Craton, India								
Haraholli Alkaline Dykes	HAD	2	23,8	83,5	—	—	814 R 810 K	14, 15
Central Australian Shield								
Todd R. Dolomite	TRD	(34)	−43	340	(15)	(7)	Lw.Cm. S	16
Lower Arumbera Sandstone	LAS	(12)	−44	324	(14)	(12)	Up.Pc. S	16
Upper Arumbera Sandstone	UAS	(57)	−47	337	(23)	(4)	Up.Pc. S	16
Hugh R. Shale	HRS	(7)	11	37	(44)	(9)	Lw.–M.Cm. S	17
Hudson Formation	HF	(17)	18	19	(7)	(14)	M.Cm. S	18
Antrim Volcanics	ARV	14	9	160	(11)	(13)	Lw.Cm. S	19
Deception Formation	DFA	(67)	−32	11	(8)	(7)	Cm. S	20
Illara Sandstone	ISA	(43)	−30	11	(5)	(11)	Cm. S	20
Tempe Formation	TFA	(26)	−36	13	(5)	(14)	Cm. S	20
Shannon Formation	SFA	(9)	−15	35	(6)	(12)	Cm. S	20
Upper Giles Creek Dolomite	GC1	(27)	−40	30	(7)	(12)	Cm. S	20
Lower Giles Creek Dolomite	GC2	(6)	−36	20	(5)	(33)	Cm. S	20
Southeast Australian Shield								
Wooltana Volcanics	WTV	10	−62	142	(24)	(10)	*c.* 850–950[c] S	21
Copley Quartzite	CQ	6	−36	10	(23)	(14)	Up.Pc. S	21
Mirinjina Tillite	MT	16	−45	346	(14)	(10)	Up.Pc. S	21
Tapley Hill Formation	TH	(12)	−39	28	(9)	(15)	Up.Pc. S	21
Angepena Formation	AF1	5	−33	344	(36)	(13)	Up.Pc. S	21
Brachina Formation	BRM	8	−33	328	(13)	(16)	Up.Pc. S	21
Bunyeroo Formation	BU	9	−7	17	(20)	(12)	Up.Pc. S	21
Aroona Dam Sandstone	ADS	(11)	36	213	(7)	—	Lw.Cm. S	22
Kangaroo Is.	KI	16	−34	15	(15)	(12)	Lw.Cm. S	23
Hawker Group	HG	15	−27	2	(12)	(11)	Cm. S	24
Billy Creek Formation	BC	11	−37	20	(11)	(14)	Lw.–M.Cm. S	24
Lower Lake Fromme Group	LLF	15	8	205	(6)	(15)	M.Cm. S	22
Upper Lake Fromme Group	ULF	5	38	206	(18)	(12)	UP.Cm.–Lw.O S	22
Basal Lake Fromme Group	LFB	10	29	206	(15)	(13)	M.Cm. S	24
Dundas Group	DDG	(8)	−23	13	—	12	M.Cm. S	25
Pantapinna Formation	PF	4	−36	29	(32)	(17)	M.–Up.Cm. S	24
Pound Quartz	PQ	—	−69	335	—	(24)	Up.Cm. or c. Sil. *op*	21, 22
West Australian Shield								
Yilgarn basic dykes	YB	4	20	102	(19)	28	*c.* 750–700[d]	26
South American Shield								
La Tinta	LT	(53)	−79	301	(17)	(5)	734 ± 24	27, 28
Piquete Complex	PC	13	−2	346	—	10	*c.* 450 P	29
Purmanarca Vil. Sandstone	PVS	(9)	61	293	—	(32)	Cm.	30
S. Tilcara Sandstone	STS	(11)	52	27	—	(23)	Cm.	30
N. Tilcara Sandstone	NTS	(11)	49	24	—	(27)	Cm.	30
Purmamarca Sandstone	PS	(12)	5	39	—	(15)	Cm.	30
Abra de Cajas	ACS	(6)	2	28	—	(50)	Cm.	30
Cambrian Red Beds	CRD	(17)	27	310	—	(14)	Cm.	31

TABLE III (continued)

Shield Formation	Abbv.[a]	No. of Sites (samples)	Lat. (°N)	Long. (°E)	K (κ)	A_{95} (α_{95})	Possible Magnetization Age	Refs.
Kalahari Shield								
Gannakouriep dykes	GD1	(13)	−22,2	66,8	—	—	550–650 > t <0,9	32
Gannakouriep dykes	GD2	(5)	36,7	352,5	(72)	(8)	542 ± 4 AA	32
Gannakouriep dykes	GD3	(6)	80,9	208,8	(23)	(14)	PEF *op*	32
Klein Karas dykes	KK	4	20	294	(5)	(35)	*mixed op*	33
Pre-Nama dykes	PN	5	85,1	227,5	(13)	(22)	PEF *op*	33
Nama Group	N1	20	60	61	(7)	(13)	553±13<t<686±32 or Cm.−O *op*	34, 35
Nama Group	N2	4	5	271	(41)	(15)	<N1	34
Nama Group	NF1	(4)	10	301	(400)	(5)	<N1	34
Nama Group	N3	6	−2	344	(25)	(14)	<N1	34
Blaubeker Group	Na	7	60	54	(3)	(43)	*c.* 580–680 *op* M	34
Blaubeker Group	NBX	3	52	353	(45)	(19)	*c* 471 *op* M[e]	34
Richtersveld Complex	RC	3	12,7	355,2	—	—	*c.* 500 *op* M	36
Stinkfontein	ST	1	−10,4	34,1	—	—	*c.* 500 *op* M	36
Kalkpunt Formation	KG2	3	1	297	(39)	(20)	*c.* 550 *op* M	37
Sijarira Group	SJ	9	2	352	(9)	(18)	Pc.−Cm. S	38
Klipheuvel Formation	KF	—	16	316	—	—	Pc.−Cm. S	39
Doornpoort Formation	DP	6	−22	245	(58)	(9)	*c.* Cm.−O *op* M	33
Congo Shield								
Nosib Group	nql	6	27	317	(12)	(20)	*in situ*, *c.* Cm *op* or	40
Nosib Group	NQ1	6	28	323	(21)	(15)	*sc*, 840±13 R<t<1 100 U	40
Nosib Group	nq2	7	−52	200	(18)	(16)	*in situ*, *c.* Cm−0 *op* or	40
Nosib Group	NQ2	7	−51	211	(25)	(14)	*sc*, *c.* 650<t<NQ1 M	40
Nosib Group	nq3	(13)	10	292	(8)	(16)	*in situ*	40
Nosib Group	NQ3	(13)	9	295	(11)	(13)	sc <*c.* 650 M	40
Chela Group	CG	—	−30	143	—	—	900–1 000 S or cm *op*	41
Gagwe Lavas	GL	28	−29	283	(5)	(13)	815 ± 14 K	42, 43
Manyovu Redbeds	MV	5	−22	297	(5)	(40)	<890 ± 32 K, s	42, 43
Bukoban intrusives	BI	16	−11	281	(5)	(19)	806±30 or ≥1 010 K	42, 43
Mbala Dolerites	MD	2	9	100	(19)	(18)	709–1 010 K, S	44, 45
Mbozi Complex	MZ	5	72	68	(20)	(18)	753±30 K bio, S	44, 46
Lake Buanji Group	LB	6	87	263	(60)	(9)	Up.Pc.−Cm. M, or PEF *op.*	44
Sabaloka Complex	SRC	(21)	83	339	(10)	(11)	PEF *op*	47, 48
Otavi Group	dc1	4	−42	157	(7)	(38)	742–770 U or Cm−O *op*	40
Otavi Group	DC1	4	−52	186	(8)	(35)	*c.* 651 *op*	40
Otavi Group	dc2	10	−55	44	(11)	(15)	<DC1 M	40
Mulden Group	DM	6	15	283	(13)	(20)	*in situ* 550–560 R	40, 49
Mulden Group	DMS	6	13	270	(18)	(16)	*sc*	
Mulden Group	DMA	(4)	−8,7	320,6	(103)	(9)	537 or 457 R c1 *op*	40, 49
West African Shield								
Plat. d'Oujeft Group	C8	4	36	52	(17)	(25)	435–465 K c1	50, 51
Plat. d'Oujeft Group	C10	7	55	30	(87)	(7)		
Morocco Lavas	ML	—	53	34	—	—	M.Cm.	52
Tassili Sandstone	TS	—	53	26	—	(6)	Cm.−O	53
Serie Lie de Vin	SLV	—	−29	63	—	(21)	*c.* C–P *op*	54
Amouslek Tuffs	AT	—	−42	70	—	(13)	*c.* C–P *op*	54
Ourzazate Volcanics	OV	—	−30	57	—	(17)	*c.* C–P *op*	54
Damaran Belt								
Damaran Granites	DG	22	27	351	(33)	(5)	450–550 R	55, 56
Damaran gneiss	DBG	3	−43,2	154,5	(78)	(14)	1925 U or *c.* 500 *op*	55, 57
Moçambique Belt								
Ntonya Ring Complex	NR	7	15	332	(1 054)	(2)	481, 486 K bio	58
Dedza Mt. Syenite	DZ	(1)	27	340	—	—	523–556 K hb	58
							630±24 R	58
Zambezi Belt								
Hook intrusives	HI	3	14	336	—	36	519±17 R, 472 K bio	59, 60
Pan-African North-east Africa								
QS E. desert dykes	ED1	25	87	304	(36)	(5)	480–527 K	61
UR E. desert dykes	ED2	14	86	185	(48)	(6)	464–497 K	61
Dokhan Volcanics	DV1	5	−36	17	(20)	(15)	*c.* 600 *op*	61
Dokhan Volcanics	DV2	1	53,5	327	(20)	(15)	472 K bio. *op*	61
Pharusian Belt								
Adma diorite	AD	13	32,5	344,7	(22)	(9)	616±11 U, *c.* 590–616	62
Tabankort rhyolite	TR	(8)	30	322	—	(7)	*c.* 590 S	62

(continued)

Notes:

[a] Abbreviations used in Fig. 7a and 7b.

[b] R—Rb-Sr, K—K-Ar, U—U-Pb zircon, AA—$^{40}Ar/^{39}Ar$, bio—biotite, mu—muscovite, hb—hornblende, cl—clay, and unless otherwise indicated the ages were derived from total rock analyses. S—age is stratigraphically related to a dated unit. M—age determined from the position of its palaeomagnetic pole in relation to a dated palaeomagnetic pole. *op*—overprint.

[c] The actual Rb-Sr isochron age reported for rock units correlative with the Wooltana Volcanics yield a 1 345 ± 30 Ma age, but a younger age of 850–950 Ma is preferred by McWilliams and McElhinny (1980), which is not based upon any radiometric determinations.

[d] The age is estimated from a hypothetical correlation of the dykes with a regional reheating event dated by Rb-Sr. The dykes themselves have not been dated.

[e] The 471 Ma age represents the average of determinations on the Damara granites that intrude and metamorphosed the Blaubeker Group.

References:

1. McQueen *et al.* (1972); 2. Zijderveld (1968); 3. Manzoni and Nanni (1977); 4. Nagata and Yama-ai (1961); 5. Klootwijk (1973); 6. Athavale *et al.* (1972); 7. McElhinny *et al.* (1978); 8. McElhinny (1970); 9. Wensink (1972); 10. Klootwijk (1975); 11. Crawford and Compston (1970); 12. Athavale *et al.* (1963); 13. Sahastradubhe and Mishra (1966); 14. Dawson and Hargraves (pers. comm.); 15. Ikramuddin and Stueber (1976); 16. Kirschvink (1978); 17. Embleton (1972); 18. Luck (1972); 19. McElhinny and Luck (1970); 20. Klootwijk (1980c); 21. McWilliams and McElhinny (1980); 22. Embleton and Giddings (1974); 23. Klootwijk (1980b); 24. Klootwijk (1980a); 25. Giddings and Embleton (1974); 26. Giddings (1976); 27. Valencio *et al.* (1980); 28. Bonhomme and Cingolani (1978); 29. Filho *et al.* (1986); 30. Thompson (1973); 31. Creer (1965); 32. this study; 33. Piper (1975a); 34. Kröner *et al.* (1980); 35. Allsopp *et al.* (1979); 36. Kröner, pers. comm. (1985); 37. Briden *et al.* (1979); 38. Reid (1968); 39. Creer (1973); 40. McWilliams and Kröner (1981); 41. Kröner and Correia (1981); 42. Piper (1972); 43. Cahen and Snelling (1974); 44. Piper (1975b); 45. Schandelmeir (1981); 46. Cahen and Snelling (1966); 47. Briden (1973); 48. Saradeth *et al.* (1985); 49. Clauer and Kröner (1979); 50. Morris and Carmichael (1978); 51. Clauer (1981); 52. Helsley (1965); 53. Ileana (1971); 54. Hailwood (1972); 55. Corner and Henthorn (1978); 56. Kröner and Hawkesworth (1977); 57. Jacob and Burger (1978); 58. Briden (1967); 59. Brock (1967); 60. Snelling *et al.* (1961); 61. Davies *et al.* (1980); 62. Morel (1981).

to date the Lake Buanji redbeds of Tanzania (Piper, 1975b) and the Sabaloka Complex (Briden, 1973) of the Sudan because of the similarity in the position of their palaeomagnetic poles (LB and SRS, respectively in Fig. 7). Given that the Lake Buanji samples were never thermally demagnetized above 400 °C and the Devonian ages and new palaeomagnetic results published for the Sabaloka Complex (Barth *et al.*, 1983; Saradeth *et al.*, 1985), it is highly probable that all three poles represent present-day overprints (PD? in Fig. 7).

The south-easterly, positive inclination component (Table I) found in both dykes using converging remagnetization circles, has an unblocking temperature ranging between 50 and 580 °C and may be carried by fine-grained, even sub-microscopic, magnetite inclusions present in the well-preserved pyroxenes. Similar occurrences have been reported in gabbroic rocks (Evans and McElhinny, 1968), where single domain, magnetite inclusions in pyroxenes carry the primary remanence, since they apparently form at high temperatures (Fleet *et al.*, 1980). In this particular case, the pyroxenes may have shielded the magnetite inclusions from the metasomatic reactions that led to the destruction of the exposed titanomagnetite. Assuming this component is primary, then its age, as is the age of the Gannakouriep dyke swarm, is also poorly constrained, lying between 900 and 600 Ma.

Despite the large uncertainty in the age of the mafic Gannakouriep dyke swarm, its palaeomagnetic pole (GD1 in Fig. 7) corresponding to the possible primary component (Table I) does not obviously agree with previously published versions of the African APW path for this time period (Kröner *et al.*, 1980; McWilliams and Kröner, 1981; and McWilliams, 1981), which is indicated by the heavy line in Fig. 7. When the Gondwanaland continents and their corresponding 850 to 650 Ma poles (Table III) are restored to a Smith and Hallam (1970) fit, GD1 lies within Antarctica among a small cluster of poles, which include the Wooltana Volcanics of the Gawler Craton (850? Ma, WTV in Fig. 7) and the LaTinta of the South American Shield (734 Ma, LT in Fig. 7). Although this cluster does lie near the 700 Ma portions of proposed Gondwanaland, APW paths (McElhinny and Embleton, 1976; and Klootwijk, 1980), acceptance of this correlation is predicated on the assumption that another cluster of poles formed by the Gagwe lavas of the Tanzanian Shield (815 Ma, GL in Fig. 7), the Yilgarn B Dykes of Western Australia Shield (~750 Ma, YB in Fig. 7), and the Malani Rhyolites of the Aravalli Craton (729 Ma, MR1 in Fig. 7) are significantly older in age. This is certainly not obvious from the available radiometric data. Furthermore, such a correlation would imply that the GD1 pole was actually younger than the palaeomagnetic pole derived from the overlying Nama Group (N1 in Fig. 7), although N1 and other poles from southern African sediments (McWilliams and Kröner, 1981) with similar orientations may represent Cambrian–Ordovician overprints. As this time period (850–650 Ma) is critical in consideration of the evolution of the Pan-African mobile belts, high quality, more well-dated palaeomagnetic poles of this vintage will be required before any conclusions can be reached.

VII. CONCLUSIONS

Palaeomagnetic and $^{40}Ar/^{39}Ar$ analyses of two dykes from the Gannakouriep dyke swarm indicate the swarm has been overprinted by younger metamorphic/thermal events that have disturbed both the palaeomagnetic and argon isotopic systems. Three main results may be summarized as follows:

1. one dyke (8212) yields a secondary component carried by pyrrhotite, consistent in direction with other palaeomagnetic poles of Pan-African age. $^{40}Ar/^{39}Ar$ analyses recovers a Pan-African overprint age for the same dyke. This date (542 ± 4 Ma), therefore, provides a good estimate of the timing of remagnetization;
2. the other dyke (8215) yields a secondary component which is parallel to the present axial dipole field and is a viscous overprint carried by fine-grained hematite. $^{40}Ar/^{39}Ar$ analyses yield a saddle-shaped spectra, suggesting partial loss of radiogenic argon during a younger event, quite possibly during the Karoo magmatic episode; and
3. both dykes yield a magnetization carried by magnetite that may be primary, dating back to the intrusion of the dykes between 600 and 900 Ma. The pole corresponding to this vector does not obviously agree with published versions of the African APW paths, but it does lie on published versions of the Gondwanaland APW paths.

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APPENDIX
Palaeomagnetic Data from the Gannakouriep Dyke Swarm Sites

Sample No.	N	Dec.	Inc.	Steps	M.A.D.	Comp.
821201.01	7	0,4	8,6	N–17,5	12,7	N−, NW+
	6*	339,5	22,3	17,5–82,5	11,7	NW+
	12	251,4	15,1	N–82,5	0,7	(N−, NW+)
821201.02	4*	346,6	14,7	N–400	7,5	NW+
	6	35,3	−39,3	400–660	15,5	NE−?
	9	89,8	35,6	N–660	0,3	(NW+, NE−)
821202.01	4*	343,8	37,7	5–17,5	12,4	NW+
821202.02	5	1,9	2,2	N–350	21,5	N−, NW+
	12	348,6	1,4	350–600	27,5	N−, NW+
	16	265,4	3,6	N–600	1,7	(N−, NW+)
821203.01	5	330,2	26,2	2,5–12,5	14,4	N−, NW+
	7*	344,4	43,1	12,5–50	11,3	NW+
	12	245,0	4,9	N–50	0,5	(N−, NW+)
821203.02	4*	322,3	26,1	N–350	4,8	NW+
	4	20,6	70,3	350–580	15,4	NW+, SE+
	7*	221,9	19,4	N–580	1,4	(NW+, SE+)
821204.01	13*	331,9	37,9	N–50	5,2	NW+
821204.02	5*	320,2	37,7	N–350	8,9	NW+
	7	145,0	60,4	350–600	17,0	SE+
	11*	234,6	−0,6	N–600	1,9	(NW+, SE+)
821205.01	5	11,8	6,8	N–10	11,6	N−, NW+
	5*	329,4	42,2	10–25	8,8	NW+
	9	111,0	38,5	N–25	0,7	(N−, NW+)
821205.02	5*	337,1	19,3	N–350	17,1	NW+
	5	310,2	18,5	350–500	15,8	NW+, SE+
	4	54,1	54,3	500–580	14,4	NW+, SE+
	9	247,3	−5,6	N–500	32,4	(NW+, SE+)
	8*	203,8	32,4	350–580	0,5	(NW+, SE+)
821206.01	15*	344,0	29,9	N–50	8,5	NW+
821206.02	4*	349,1	27,7	N–540	12,7	NW+
	4	240,9	56,8	540–620	15,6	SW+?
	9	267,4	−33,1	N–620	0,4	(NW+, SW+)
821501.01	12*	44,2	−46,1	N–40	1,8	N−
821501.02	16*	47,5	−47,5	N–690	1,6	N−
821502.01	7	257,2	−28,6	N–15	15,4	W−?
	7	280,4	8,1	15–70	1,6	NW?
	13	5,4	78,6	N–70	0,6	(W−, NW+)
821502.02	17	266,1	−1,2	N–690	2,4	W−?
821502.03	19	247,7	−18,0	N–660	1,3	SW−?
	1	149,6	45,3	660–680	—	SE+
	20*			N–680	0,3	(SW−, SE+)
821503.01	5	77,2	−43,4	N–10	18,1	N−, SE+
	9*	357,5	−55,6	10–40	3,7	N−
	13	50,5	26,4	N–40	0,7	(SE+, N−)
821503.02	9*	43,3	−61,9	N–550	3,6	N−, SE+
	2	53,2	16,3	550–580	9,9	N−, SE+
	10	143,2	−4,4	N–580	0,9	(N−, SE+)
821503.03	8	40,9	−67,5	N–500	3,6	N−, SE+
	4	57,4	8,4	500–680	4,9	N−, SE+
	11*	144,9	−7,7	N–680	0,1	(N−, SE+)
821504.01	10*	352,1	−44,1	N–30	10,1	N−
	4	2,1	−28,9	30–60	7,2	N−, SE+
	13	99,1	−11,8	N–60	0,2	(N−, SE+)
821504.02	9*	348,9	−31,7	N–520	5,4	N−
	3	21,9	41,9	520–680	2,1	N−, SE+
	11*	90,5	−19,8	N–680	0,2	(N−, SE+)
821505.01	10*	350,4	−60,4	5–35	5,3	N−
	3	355,7	18,4	35–60	20,7	N−, SE+
	12	89,8	2,3	5–60	0,4	(N−, SE+)
821505,02	8*	352,0	−60,2	N–450	5,7	N−
	4	16,2	13,9	450–560	8,0	N−, SE+
	11*	284,4	14,9	N–560	0,2	(N−, SE+)
821506.01	15*	342,6	−51,7	N–60	6,4	N−
	2	25,3	17,8	60–70	10,8	N−, SE+
	7	19,9	25,7	70–500	6,3	N−, SE+†
	16	109,2	−23,4	N–70	0,1	(N−, SE+)
821506.02	10*	359,3	−54,6	N–520	7,9	N−
	3	98,4	39,4	520–600	19,2	SE+
	12*	295,1	29,8	N–600	0,5	(N−, SE+)
821507.01	16*	336,8	−67,6	N–70	3,4	N−
821507.02	17*	337,0	−59,7	N–690	6,3	N−

*Sample direction included in average site statistics.
†821506.01 was thermally demagnetized) from 200 to 680 °C after A.F. demagnetization up to 70 mT.

Note:
n — number of demagnetization steps included in the least squares line or plane fit. Anomalous directions were excluded.
Dec., Inc. — the orientation of the least squares line or perpendicular to the least squares remagnetization plane.
Steps — the range of demagnetization fields (2,5–130 mT for A.F. demagnetization) or temperatures (200–680 °C for thermal demagnetization) included in the least squares fit.
M.A.D. — mean angular deviation as defined by Kirschvink (1978).
Comp. — classification used to describe component.

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